

C-ION Advanced Oxidation System

Advanced Oxidation and Non-Thermal Plasma Contaminant Removal.

C-ION

Technical Specifications

C-ION technical description	Parameter
Advanced oxidation process	Technology type
Electrical glow discharge	Main process
Non-thermal plasma, NTP	Plasma type
Reactive oxygen species, ROS	Main oxidants
Oxidation and breakdown of difficult organic and inorganic contaminants	Primary function
Electrical energy creates plasma discharge in the treatment chamber	Operating principle
Trace-organic removal, arsenic removal, manganese removal and bulking-sludge control	Main applications
Iron coagulation and C-MEM ultrafiltration	Optional treatment
C-MEM immersed hollow-fibre UF	Main membrane option
May require iron coagulant for arsenic and manganese removal	Chemical requirement
Not a biological reactor	Biological process
Not required as the primary C-ION process	Ion exchange resin
Not applicable	Brine regeneration
Oxidation products, oxidized metals and chemical sludge where iron coagulation is used	Main by-products